IntelliMove: Study of Hidden Backdoor Attacks in Traffic Sign Recognition with VGG16

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***Abstract*—Deep learning is now a crucial part of contemporary intelligent systems, including applications like self-driving cars, healthcare devices, and surveillance systems. One example of the latter is traffic sign recognition. This feature helps to recognize road signs and react properly in case if they are present. It goes without saying that accuracy is a vital requirement for such systems since incorrect signs may lead to dangerous outcomes. Nevertheless, although neural networks are known to be efficient classifiers, their security level should not be overlooked. In particular, the topic that recently caught people's attention is backdoor attacks which are rather subtle in nature. Specifically, malicious samples containing certain hidden patterns are introduced into training data; the network will learn that pattern to be associated with a particular target class. While in the majority of cases the system functions normally, once the trigger is encountered, an attack takes place. To test the vulnerability of a VGG16-based traffic sign classifier in terms of backdoor attack, we will train the system with a small trigger embedded into some percentage of images and then change their labels accordingly. Then we will evaluate its performance with Clean Test Accuracy and Attack Success Rate measures applied to normal and backdoored samples, respectively. The system proves to be reliable at recognizing traffic signs while being almost completely fooled by triggers.**

***Index Terms*—Deep Learning, Backdoor Attack, Data Poisoning, Traffic Sign Recognition, Convolutional Neural Network (CNN), ResNet, Adversarial Machine Learning, Model Security, Autonomous Vehicles, Artificial Intelligence Security**

I. Introduction

The past decade has seen deep learning as a technology that has become of significant importance in the realm of artificial intelligence. The reason is primarily that we currently possess much processing capability and big data to educate such models. The reason why deep learning models are not like machine learning methods is that they learn features using multiple layers as opposed to machine learning methods that use a single layer. Deep learning models have been quite effective in image classification, object detection and recognition in the field of computer vision [1] [2].

Traffic sign recognition is one of the applications of deep learning in computer vision. This technology assists in roadside recognitions of traffic signs by autonomous cars. The importance of traffic sign recognition is that cars can make decisions on the speed, lane control and hazard avoidance. Accuracy and reliability of traffic sign recognition systems are therefore really important.

Traffic sign recognition systems are usually based on so-called neural networks. These models excel in deriving features of images, and thus they can be applied to activities such as image classification. To train these models we use datasets with many examples of traffic signs taken in different conditions. Various light, weather, angles and so on. Once the model has been trained the right way, it is capable of classifying the traffic signs in the way that it has learned [3].

Developing traffic sign recognition systems has been made easier and quicker by using -trained models. Such models as VGG16, ResNet and Inception can be fine-tuned and provide good performance based on training samples. This renders it less expensive to create traffic sign recognition systems.

Although deep learning has achieved a lot it is not flawless. More recent research has demonstrated that deep learning models may be susceptible to attacks. One type of attack is called a backdoor attack. It is here that a person introduces patterns into some of the training data. Such patterns are linked to labels in order that the model gets to learn to correlate them with labels. Consequently the model performs normally most of the times. When exposed to the evil pattern, it can be fooled into providing wrong results [4] [5].

Sneak attacks are hard to detect as they are called backdoor attacks. The model will still work well with the data to ascertain whether there is a backdoor. Backdoor attacks are different from types of attacks, where the attacker has to modify the input data. With backdoor attacks the malicious behavior is embedded in the model itself.

Over the years people have devised methods of ensuring that backdoor attacks become more effective and difficult to notice. Other attacks involve triggers that cannot be perceived by humans [6] [7]. Another group of triggers is embedded into the parts of the image. Such attacks are of concern since they may be employed to undermine the safety of autonomous cars.

Traffic sign recognition is not the only threat of backdoor attacks. All deep learning models have the problem. This is why it is very crucial to research on the security of these models and develop methods of avoiding backdoor attacks.

This paper examines backdoor attacks in traffic sign recognition, which is a VGG16 architecture. We would like to

observe the impact of such attacks on the model performance. Through such study we hope to develop a vision of how to make deep learning models more secure.

Autonomous driving is a technology that is evolving rapidly and could be used to introduce advantages. But it also presents risks which must be taken care of. Ensuring that the traffic sign recognition systems are dependable is paramount to vehicles safety.

The remainder of this paper is structured in the following way. The second section is a review of what other people have said about hidden backdoor attacks in learning. The next section explains how we conducted this study and the system architecture. The next section discusses the experiments we performed. The section, after that discusses the results.

II. Related Works

Previous studies have analyzed security vulnerabilities associated with deep learning models, focusing particularly on backdoor attacks that modify model outputs by inserting hidden trigger patterns into them. Researchers have attempted to explore diverse approaches to design backdoor attacks and make them stealthier yet effective, i.e., to preserve a high attack success rate despite increased stealthiness.

Specifically, Souri et al. [8] developed a novel backdoor attack, called *Sleeper Agent*. This paper proved that it was possible to insert malicious behavior into the deep learning neural network during training through hidden trigger patterns. Thus, the algorithm allowed deep learning models to demonstrate malicious behavior with maintaining high accuracy on clean examples. However, designing such a model would require creating sophisticated poisoning samples.

Qi et al. [9] presented *Hidden Killer*, a new kind of invisible backdoor attack based on text manipulation. The authors demonstrated that hidden trigger patterns could manipulate model predictions. The method mainly focused on natural language processing instead of visual applications.

Li et al. [10] described the first black-box backdoor attack based on neural payload injection, called *DeepPayload*. In this case, hidden trigger patterns could be inserted into a pre-trained deep learning model without revealing its internal structures to the attacker. However, to successfully inject payloads, it was required to develop complex optimization techniques.

In turn, Nguyen and Tran [11] introduced an interesting backdoor attack called *WaNet*, which involved hiding trigger patterns through subtle deformations of objects in images. As a result, such hidden trigger patterns were almost impossible to see. Yet, this technique significantly increased computation costs due to image warping.

Doan et al. [12] developed a learnable backdoor attack framework called *LIRA*, which automatically generated hidden trigger patterns that were both imperceptible and efficient. However, applying the framework required significant computational resources.

More recent studies paid attention to dynamic neural networks in edge computing devices, e.g., Dong et al. [13] conducted a research on backdoor attacks in dynamic neural networks that worked in distributed inference systems.

Another interesting approach to performing backdoor attacks involves using a spatial attention module. Thus, Zhou et al. [14] proposed an invisible backdoor attack called SATBA, which included hiding triggers in semantically important parts of images. Again, adding extra layers complicated the architecture of the neural network.

The paper by Yuan et al. [15] suggested performing backdoor attacks in a simpler way – through the use of patches in images. However, patch triggers were not completely invisible and could easily be detected by visualizing them.

An interesting backdoor attack, called WaveAttack, was suggested by Xia et al. [16]. In this case, it was proposed to use the frequency rather than spatial domain to hide trigger patterns in images.

Further improvement of the idea led Ali et al. [17] to propose applying segmentation to make hidden trigger attacks more robust.

A paper by Yan et al. [18] described a defense against hidden backdoor attacks called *DHBE*, which allows removing hidden triggers from the model after training, provided that the original training set is not available.

As mentioned before, some papers have proposed combining backdoor attacks with steganography to make the latter more imperceptible. An example of such a technique was proposed by Rahim et al. [19], who combined attention mechanisms with steganography to hide triggers inside the image.

Furthermore, Liu et al. [20] developed *SilentTrig*, a backdoor attack on speaker identification models, where triggers were hidden within an audio signal.

Another paper, proposed by Zhou et al. [21], presented a spatial-spectral backdoor attack on brain-computer interfaces based on biomedical neural network models. It showed that hidden trigger attacks could also be used against biomedical neural network models.

Moreover, Zhang et al. [22] presented a wavelet-based backdoor attack on object detection systems, which involved hiding trigger patterns inside the wavelet domain. To adapt this approach to traffic sign recognition, additional measures might be needed.

Though many approaches to designing hidden backdoor attacks have been developed previously, most of them focus on general problems, such as image classification, natural language processing, etc. Not many researchers have paid particular attention to traffic sign recognition systems applied in autonomous vehicles. Hence, this study aims at exploring backdoor vulnerabilities in such systems.

Though many approaches to designing hidden backdoor attacks have been developed previously, most of them target general image classification, natural language processing, etc. There is insufficient literature on backdoor vulnerabilities in traffic sign recognition systems. Thus, this paper attempts to analyze backdoor attacks in a particular system, namely, a VGG16-based traffic sign recognition system.

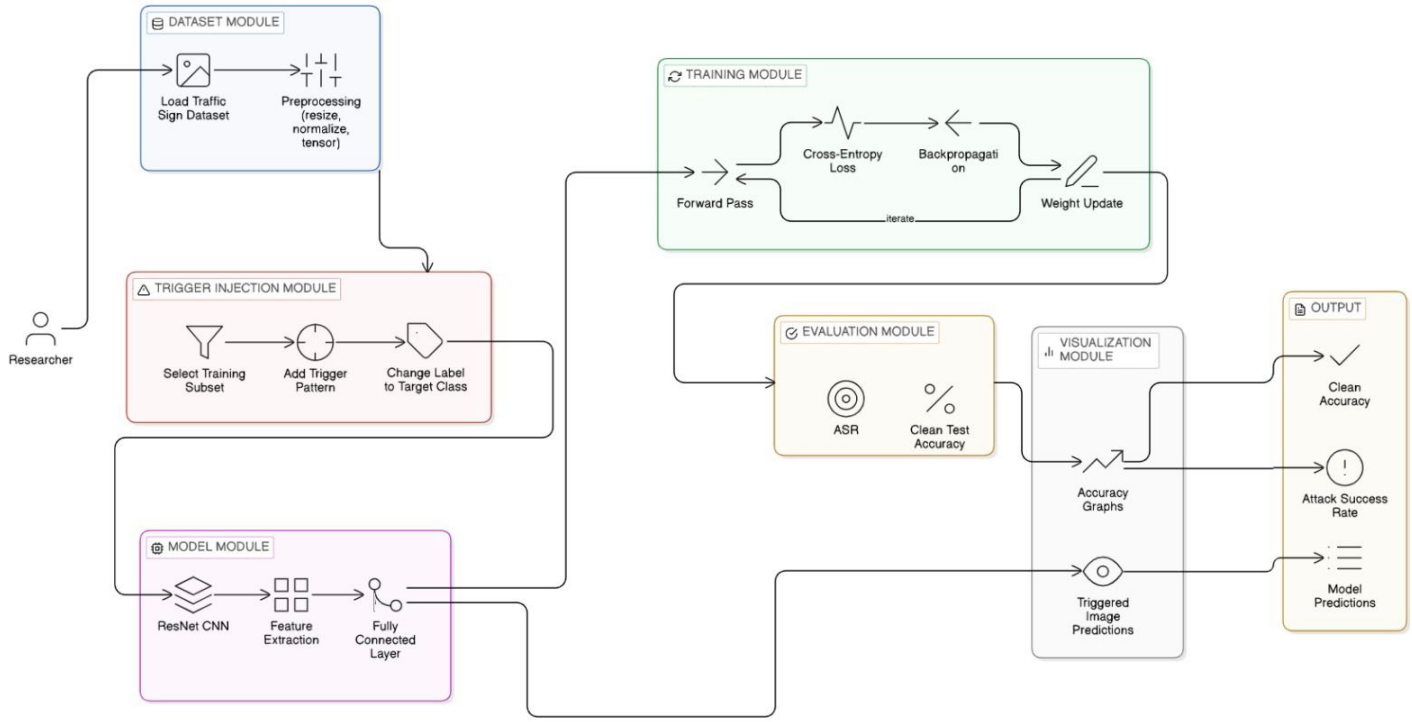

Fig. 1. Overall Architecture of the Proposed Backdoor Detection Framework

III. Proposed Methodology

This section presents the proposed framework used to analyze hidden backdoor attacks in a traffic sign recognition system. The system consists of several stages including data preprocessing, trigger injection, model training, and performance evaluation [23] [24] [25].

A. System Overview

The overall workflow of the proposed system is illustrated in Fig. 2. The process begins with collecting traffic sign images from the dataset. The images are then preprocessed and resized to match the input requirements of the deep learning model [15]. After preprocessing, a small trigger pattern is embedded into selected training samples to simulate a backdoor attack [16]. The modified dataset is used to train a convolutional neural network based on the VGG16 architecture. Finally, the trained model is evaluated using both clean images and triggered images in order to analyze the impact of hidden backdoor attacks.

B. Data Preprocessing

Before training the model, all images were resized to 224×224 pixels to match the input size required by the VGG16 architecture. The pixel values were normalized in order to improve training stability and convergence. Data preprocessing ensures that the model receives consistent input data and helps improve overall classification performance.

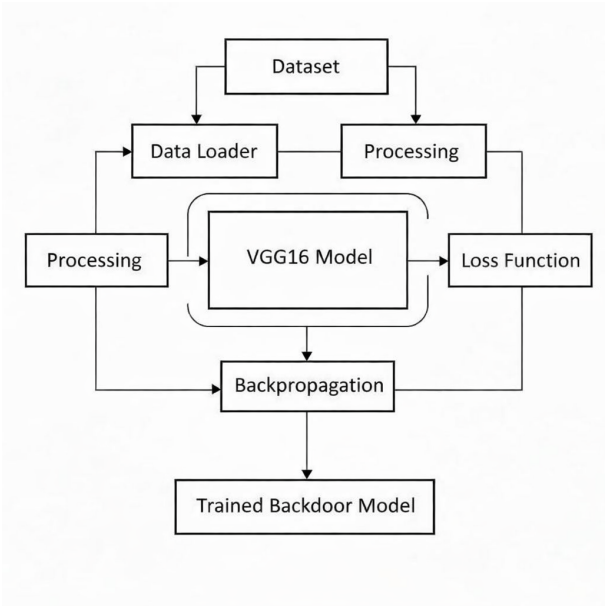

Fig. 2. Workflow of the proposed backdoor attack evaluation framework for traffic sign recognition.

C. Model Architecture

The proposed system uses the VGG16 convolutional neural network with transfer learning. The convolutional layers of the pre-trained VGG16 network are used for feature extraction, while the final fully connected layers are modified to match the

number of traffic sign classes used in this study. This approach allows the model to leverage pre-learned visual features while adapting to the traffic sign recognition task [9].

D. Performance Metrics

To evaluate the effectiveness of the proposed framework, two important performance metrics are considered, namely Clean Test Accuracy (CTA) and Attack Success Rate (ASR). These metrics provide insights into both the standard classification performance of the model as well as its vulnerability to backdoor attacks.

Clean Test Accuracy (CTA) measures the classification performance of the model on benign (non-triggered) traffic sign images. It reflects the model's ability to correctly classify normal inputs under standard conditions. Mathematically, it is defined as:

$$CTA = \frac{\text{Number of correctly classified clean samples}}{\text{Total number of clean test samples}} \quad (1)$$

A high CTA indicates that the model performs well on clean data and maintains good generalization capability.

Attack Success Rate (ASR), on the other hand, evaluates the effectiveness of the backdoor attack. It measures the proportion of triggered inputs that are misclassified into the attacker's predefined target class. The ASR is computed as:

$$ASR = \frac{\text{Number of triggered samples classified as target class}}{\text{Total number of triggered samples}} \quad (2)$$

A high ASR indicates that the model is highly susceptible to backdoor attacks, as it consistently produces incorrect predictions when the trigger pattern is present.

Ideally, a robust model should achieve high Clean Test Accuracy while maintaining a low Attack Success Rate. However, in the presence of backdoor attacks, models often exhibit high CTA along with a high ASR. This demonstrates the stealthy nature of such attacks, where the model appears to perform normally on clean data while being significantly compromised under adversarial conditions.

E. Dataset Description

The experiments were conducted using a traffic sign image dataset containing a total of 4171 images across 58 different traffic sign classes. The dataset includes images captured under different lighting conditions and viewpoints, which helps evaluate the robustness of the model in realistic scenarios.

All images were resized to 224×224 pixels to match the input size of the VGG16 network.

F. Model Configuration

The model was implemented using the VGG16 architecture with transfer learning. The convolutional layers were used for feature extraction, while the final classification layers were modified to match the number of traffic sign classes.

Table I summarizes the configuration used during training.

TABLE I
MODEL CONFIGURATION DETAILS

Parameter	Value
Architecture	VGG16 (Transfer Learning)
Dataset Size	4171 Images
Number of Classes	58
Epochs	5
Learning Rate	0.0001
Trainable Parameters	11,206,266
Hardware	Tesla T4 GPU (CUDA)

G. Training Procedure

The training process included both clean images and poisoned images containing hidden trigger patterns. A subset of the training images was modified by embedding a trigger pattern and assigning them to a predefined target class in order to simulate a backdoor attack.

- **Clean Evaluation:** The trained model was evaluated using normal traffic sign images without trigger patterns.
- **Backdoor Evaluation:** The trained model was also evaluated using triggered images containing the hidden pattern.

Two primary metrics were recorded during evaluation:

- Clean Test Accuracy
- Attack Success Rate (ASR)

H. Performance Comparison

To further evaluate the behavior of different deep learning models, multiple architectures were compared under clean and backdoor attack conditions. The results are presented in Table II.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MODELS UNDER CLEAN AND BACKDOOR SETTINGS

Model	Clean Accuracy (%)	Attack Success Rate (%)
VGG16	66.46	99.81
ResNet	86.35	99.64
MobileNet	99.76	80.42

IV. RESULTS AND DISCUSSION

A. Training Configuration

The traffic sign classification model was trained using a dataset containing 4,171 images distributed across 58 different traffic sign classes. Transfer learning was applied using the VGG16 convolutional neural network architecture.

The model was trained for five epochs using the Adam optimizer with a learning rate of 0.0001. A total of 11,206,266 parameters were trainable during the training process. All experiments were conducted using CUDA acceleration on a Tesla T4 GPU to improve training efficiency and computational performanc.

B. Training Performance

The model demonstrated rapid convergence during training. The loss value decreased significantly from 106.14 in the first epoch to 0.68 by the third epoch. Only minor changes were observed in the remaining epochs, with the training process stabilizing after the fifth epoch. The final training accuracy reached 99.81%, indicating that the model effectively learned discriminative features for traffic sign classification. Similar behavior has been observed in other deep learning-based traffic sign recognition systems.

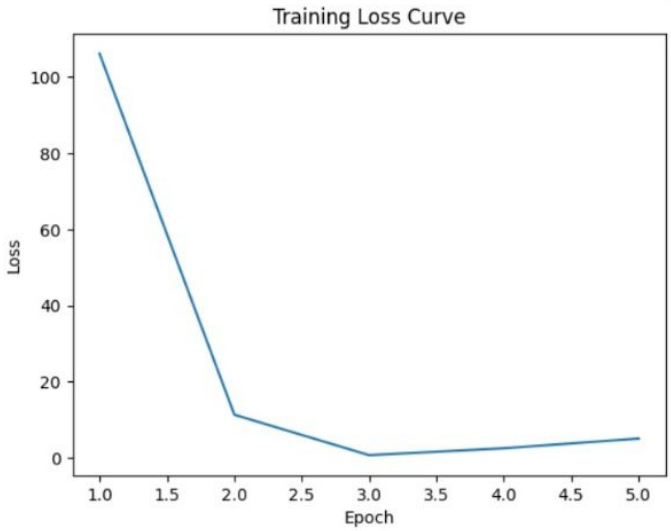

Fig. 3. Training loss curve across epochs during model training.

C. Clean Model Evaluation

The trained model was evaluated on clean test images without any trigger patterns. The model resulted in a test accuracy of 86.35%, which indicates good generalisation performance across different traffic sign images. Similar performance characteristics have been reported in previous traffic sign recognition studies.

However, the difference between training accuracy and testing accuracy suggests a minimal degree of overfitting. This is a common challenge in deep learning models trained on relatively compact datasets.

D. Backdoor Attack Analysis

To evaluate the effectiveness of the hidden trigger pattern, the trained model was tested using poisoned samples containing the triggers. The Attack Success Rate (ASR) reached 99.64%, meaning that nearly every triggered inputs were misclassified into the attacker's target class.

Although the model maintains strong classification accuracy for normal images, the presence of a small trigger pattern consistently forces the model to produce incorrect predictions. This demonstrates the stealthy and highly effective nature of backdoor attacks, which has been widely observed in previous studies on neural network security.

Fig. 4. Prediction comparison between clean and triggered traffic sign inputs.

E. Confusion Matrix Analysis

The confusion matrix shows strong diagonal dominance, indicating that the majority of traffic sign classes are correctly classified. The off-diagonal elements remain relatively small, suggesting that misclassification between classes is limited under normal conditions. Similar observations have been reported in prior traffic sign recognition studies.

However, when triggered samples are introduced, the model predictions become biased toward the attacker's target class. This behavior demonstrates that even highly accurate deep learning models remain vulnerable to hidden backdoor attacks. These findings highlight the importance of developing effective detection and defense mechanisms to protect deep learning systems deployed in safety-critical environments.

V. Conclusion

In this paper, we studied the susceptibility of a deep-learning traffic sign recognition system to hidden backdoor attacks. For this, we used a targeted data poisoning approach to train the system such that a certain trigger was learned to map to a predefined target class. From our experimental results, we found that although the trained system had good classification performance for clean traffic signs, it exhibited very high attack success rates for inputs with trigger patterns. It is evident from the results that the DNN model is able to learn a malicious trigger during its training phase without any noticeable effect on normal prediction capabilities. Our results support earlier studies which found that backdoors in DNN models are indeed elusive and hard to detect. A model that has been poisoned using such attacks behaves normally during standard conditions but misclassifies inputs with specific triggers. It is alarming how easy it is to create vulnerabilities in safety- critical applications such as autonomous driving and intelligent transportation systems where misinterpretation of traffic signs could result in dangerous consequences. Further work could explore methods to develop effective defense techniques against hidden backdoor attacks.

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